

Effects of Dual Task Training on a Child with Cerebral Palsy: A Case Report Using ICF-Based Quantitative Analysis of Functional Progress. Raphael Sadiwa (San Antonio Independent School District, San Antonio, TX).

Disclosure: None declared.

Objective: Dual-task training combining cognitive and postural activities has been extensively studied as an intervention among elderly patients with balance impairment. However, its effects on pediatric patients with balance impairment are unknown. This case report examines the effect on functional outcomes of dual-task training in a young patient with cerebral palsy with diplegia.

Design: Case report.

Setting: Public school district.

Patient: 10-year-old boy with diplegic cerebral palsy.

Case Description: Patient is a 10-year-old ambulatory boy (without assistive device) who exhibited falling in crowded environments and during increased cadence of his gait. Intervention included the use of dual-task training with the patient being asked to stand on a compliant surface (2-inch foam) while performing mathematical calculations.

Main Outcome Measure: Progress was assessed at baseline and 8 weeks later using the School Function Assessment (SFA) Travel section. Each SFA criterion score has a corresponding standard error of measurement (SEM). Confidence intervals were analyzed based on the SFA criterion scores and SEM.

Assessment/Results: The patient made immediate improvement in his balance and functional skills in the school environment. SFA Travel criterion score changed from 65, SEM 3 (95% CI: 59.12–70.88) to 77, SEM 3 (95% CI: 71.12–82.88).

Conclusions: The use of dual-task training was beneficial in improving the skills of this young patient. This intervention may present as a cost-effective way of addressing balance impairment among children with cerebral palsy. Dual-task training can be easily incorporated into pediatric rehabilitation programs.

Key Words: Cerebral palsy; Rehabilitation.